

Brian Kim

📍 College Park, MD

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Education

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| <p>PhD University of California, Los Angeles, Statistics</p> <ul style="list-style-type: none"> Dissertation: Population Size Estimation using Multiple Respondent-Driven Samples | <p>Los Angeles, CA
Sept 2012 – Oct 2017</p> |
| <p>BA Amherst College, Mathematics & Philosophy</p> <ul style="list-style-type: none"> Graduated with Honors in Mathematics Thesis: Using Mode Identification Clustering Methods with Pitch F/X Data | <p>Amherst, MA
Sept 2008 – June 2012</p> |

Research Interests

- Data Science Education; Applications of AI and Machine Learning in Social Science and Survey Methodology; Social Network Analysis; Network Sampling; Respondent-Driven Sampling; Population Size Estimation

Experience

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| <p>College of Behavioral and Social Sciences (BSOS), University of Maryland, Co-Director, Social Data Science Major</p> <ul style="list-style-type: none"> Led development of a new interdisciplinary major in Social Data Science and acted as the founding BSOS co-director, including managing the budget, setting up infrastructure, and working with partner units on campus. Launched and oversaw innovative pedagogical projects, such as a "sidecar" course program in which students take a supplementary 1-credit course alongside existing statistics and data science courses based on performance on a pre-test. Developed computational infrastructure with a cloud computing environment and continually manage and oversee the computing environment to support students within BSOS. Led a team of students and staff members to build a data repository hosting classroom-ready datasets and providing resources to support instructors; and launched the undergraduate Data Curation Fellowship in which students identify, obtain, clean, and curate datasets. Served as chief data scientist for the ongoing campus-wide internal BEACON (Building Early Alerts for Campus Outreach Needs) project, which used student data from the Registrar, Dean of Students Office, campus surveys, and learning management systems with machine learning techniques to predict student outcomes, identify opportunities for intervention, and provide actionable insight for advisors. | <p>College Park, MD
Sept 2021 – present
5 years 2 months</p> |
| <p>Social Data Science; Joint Program in Survey Methodology, Associate Research Professor</p> <ul style="list-style-type: none"> Taught classes on introductory statistics, data science, machine learning, AI, natural language processing, and other topics for both undergraduate and graduate students, incorporating innovative pedagogical techniques and utilizing traditional and flipped classroom formats in online, in-person, and HyFlex modes. Built and taught new courses for the Social Data Science major and the Survey and Data Science graduate program, such as SDSB233: Data Science for Social Science, SDSB326: Python Programming for Social Science, SURV673: Introduction to Python and SQL, and SDSB123: Social Data Science: Pathways and Applications. Advised and mentored students; engaged graduate and undergraduate students in collaborative research and publications. Wrote grant proposals and executed research plans. | <p>College Park, MD
Oct 2017 – present
9 years 1 month</p> |

Joint Program in Survey Methodology (JPSM), University of Maryland, Director of Graduate Studies

- Oversaw updates to the Master's in Survey and Data Science program, such as incorporating neural networks and LLMs into course material.
- Assisted graduate students in finding funding through grants or assistantship opportunities on campus.
- Chaired the Master's Admissions committee and managed recruitment efforts.

College Park, MD
Sept 2024 – Aug 2026
2 years

The Coleridge Initiative, Consultant/Facilitator

- Provided guidance as a facilitator with the Applied Data Analytics program for teaching data science and programming topics to working professionals in public policy.
- Led workshops on record linkage, text analysis, network analysis, and machine learning.
- Consulted on NSF NCSES projects using Survey of Earned Doctorates, Survey of Doctorate Recipients, Higher Education Research and Development, and IPEDS data.
- Developed modularized lessons and notebooks for introductory statistics and programming in Python, R and SQL.

Sept 2020 – Jan 2022
1 year 5 months

UCLA Institute for Digital Research and Education, Statistical Consultant

- Assisted graduate students and professors in cleaning, merging, and preparing data for analysis using R, Python, Stata, SPSS, and other statistical software.
- Helped clients with their dissertations, publications, and other projects using methods such as mixed modeling, principal components analysis, simulation, and more.

Los Angeles, CA
May 2014 – June 2016
2 years 2 months

Teaching

University of Maryland, Instructor (Undergraduate Courses)

- SDSB180: **Introduction to Data Management** (Fall 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026)
- SDSB123: **Social Data Science: Pathways and Applications** (Fall 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026)
- SDSB326: **Python Programming for Social Science** (Spring 2024)
- SDSB233: **Data Science for Social Sciences** (Fall 2019, Spring 2020, Spring 2022, Fall 2022, Spring 2023)
- INST 314: **Statistics for Information Science** (Fall 2018)
- INST 354: **Decision Making for Information Science** (Spring 2019)

Oct 2017 – present

University of Maryland, Instructor (Graduate Courses)

- SURV 613: **Machine Learning and AI for Social Sciences** (Fall 2020, Fall 2021, Spring 2023, Spring 2024, Spring 2025, Spring 2026)
- SURV 622: **Fundamentals of Data Collection** (Spring 2020, Spring 2021, Spring 2022)
- SURV 673: **Introduction to Python & SQL** (Fall 2018, Spring 2019, Summer 2019, Fall 2019, Fall 2020, Summer 2020, Summer 2021)
- SURV 699M: **Review of Statistical Concepts** (Summer 2018, Summer 2019, Summer 2020, Summer 2021)
- SURV 701: **Analysis of Complex Sample Survey Data** (Spring 2021)
- SURV 727: **Fundamentals of Computing and Data Display** (Fall 2022, Fall 2023, Fall 2024, Fall 2025)
- SURV 736: **Introduction to Web Scraping with R** (Summer 2025)

Oct 2017 – present

University of California, Los Angeles, Instructor

- Stat 98T: **Six Degrees of Separation: Studying the World Through Social Networks** (Winter 2017)

Jan 2017 – Mar 2017

University of California, Los Angeles, Teaching Assistant

- Stat 10: **Introduction to Statistical Reasoning** (Fall 2013, Winter 2014, Spring 2014, Fall 2014, Winter 2015, Spring 2015, Fall 2015, Winter 2016, Spring 2016)

Sept 2013 – June 2016

Publications

Using a Stopping Rule to Optimize Cost-Quality Tradeoffs in a Large, Mixed-Mode Survey: A Simulation Study

J. Wagner, B. West, **B. Kim**, D. Suolang, C. Engstrom, J. Sinibaldi
www.doi.org/10.1177/0282423X241287452

Hidden Population Size Estimation and Diagnostics Using Two Respondent-Driven Samples with Applications in Armenia

B. Kim, L. G. Johnston, T. Grigoryan, A. Papoyan, S. Grigoryan, K. R. McLaughlin
www.doi.org/10.1002/bimj.202200136

Using Geographical Data and Rolling Statistics for Diagnostics of Respondent-Driven Sampling

B. Kim, M. Ogwal, E. Sande, H. Kiyangi, D. Serwadda, W. Hladik
www.doi.org/10.1016/j.socnet.2020.05.001

Population Size Estimation Using Multiple Respondent-Driven Sampling Surveys

B. Kim, M. Handcock
www.doi.org/10.1093/jssam/smz055

Easy-to-Use Cloud Computing for Teaching Data Science

B. Kim, G. Henke
www.doi.org/10.1080/10691898.2020.1860726

Partnering with a global platform to inform research and public policy making

F. Kreuter, N. Barkay, A. Bilinski, A. Bradford, S. Chiu, R. Eliat, J. Fan, T. Galili, D. Haimovich, **B. Kim**, S. LaRocca, Y. Li, K. Morris, S. Presser, T. Sarig, J. A. Salomon, K. Stewart, E. A. Stuart, R. Tibshirani
www.doi.org/10.18148/srm/2020.v14i2.7761

Software and Data

sspse: Estimating Hidden Population Size using Respondent Driven Sampling Data

M. Handcock, K. Gile, **B. Kim**, K. McLaughlin
cran.r-project.org/package=sspse

The University of Maryland social data science center global COVID-19 trends and impact survey, in partnership with Facebook

J. Fan, Y. Li, K. Stewart, A. R. Kommareddy, A. Garcia, J. O'Brien, A. Bradford, X. Deng, S. Chiu, F. Kreuter, N. Barkay, A. Bilinski, **B. Kim**, T. Galili, D. Haimovich, S. LaRocca, S. Presser, K. Morris, J. Salomon, D. Vannette
covidmap.umd.edu/api.html

Book Chapters and Non-Refereed Publications

The Value of Science: Special Theme

J. Lane, **B. Kim**, F. Kreuter, A. Nunez
[10.1162/99608f92.4a01213d](https://doi.org/10.1162/99608f92.4a01213d)

Scaling Up Data Science for the Social Sciences

B. Kim
[10.1162/99608f92.d3f14ea4](https://doi.org/10.1162/99608f92.d3f14ea4)

Workbooks

B. Kim, C. Kern, J. Morgan, C. Hunter, A. Kumar
textbook.coleridgeinitiative.org/chap-workbooks.html

Selected Conference Presentations

A Hybrid Household and Respondent-Driven Sampling Strategy for Reaching Underrepresented Groups

B. Kim, A. Gard

Machine Learning Model Selection for Complex Sample Survey Data

B. Kim

Data 8 Adoption at University of Maryland

B. Kim

Population Size Estimation Using Multiple Respondent-Driven Sampling Surveys

B. Kim, M. Handcock

Teaching Data Science Using Jupyter Notebooks and Binder

B. Kim

Assessing Respondent-Driven Sampling Using Geographical Data

B. Kim, M. Ogwal, E. Sande, H. Kiyangi, D. Serwadda, W. Hladik

Grants

Advancing Computational Expertise in Neuroscience and Psychology (co-PI)

May 2025 – May 2026

UMD Artificial Intelligence Interdisciplinary Institute: Course Development Grant

- Funding: \$10,000

Cloud-based Active Learning for BSOS Statistics and Data Science Courses (PI)

Sept 2022 – Aug 2023

UMD Teaching and Learning Transformation Center: Experiential Learning Grant

- Funding: \$135,573.72

An Evaluation of Sampling Methodologies for Reaching Underrepresented Groups (PI)

June 2022 – May 2023

University of Maryland BSOS Dean's Research Initiative

- Funding: \$19,999

Global Privacy Monitor (GPM): Learning from External Privacy Stakeholders

June 2021 – May 2024

Facebook, Inc.

- Funding: \$2,994,731

Modernizing NCSES Data Collection Approaches (co-PI)

Sept 2020 – Aug 2021

National Science Foundation, NCSES BAA

- Funding: \$88,802

Collaborative Research: Impacts of Hard/Soft Skills on STEM Workforce Trajectories (co-PI)

May 2020 – May 2025

National Science Foundation, ECR: 1956114

- Funding: \$401,716

Cooperative Agreement: Linked Dataset and Virtual Research Data Center (co-PI)

Aug 2019 – Feb 2024

National Agricultural Statistics Service

- Funding: \$653,490

Teaching Innovations Grant

June 2020 – Aug 2020

University of Maryland

- Funding: \$18,000

Teaching Innovations Grant

June 2020 – Aug 2020

University of Maryland

- Funding: \$5,000

Developing a New Course in Data Science: Introduction to Data Science for Social Sciences (PI)

July 2019 – May 2020

UMD Year of Data Science

- Funding: \$17,500

Feasibility Testing of a Respondent-Driven Sampling Approach to Recruit Human Trafficking Victims (co-PI)

Dec 2018 – July 2019

Governor's Office of Crime Control and Prevention

- Funding: \$30,000

Other Academic Work

Guest Editor

2022 – 2022

- Harvard Data Science Review, Special Theme: Value of Science. Issue 4.2.

Reviewer

2018 – present

- Journal of Statistics and Data Science Education
- Journal of Survey Statistics and Methodology
- Journal of Statistical Software
- Public Opinion Quarterly
- Harvard Data Science Review
- Computational Statistics and Data Analysis
- Journal of Official Statistics

Mentorship

Doctoral Students

Advisor

- Harold Gomes (expected 2026)

Doctoral Committee

Committee Member

- Yuting Chen, University of Maryland (2025)
- Ai Rene Ong, University of Michigan (2022)
- Jordan Epistola, University of Maryland (2022)

Graduate Students on Funded Projects

Supervisor

- Namit Shrivastava (January 2026 – May 2026)
- Ujjayini Das (September 2022 – August 2023)
- Joseph Hoskisson (September 2022 – August 2023)
- Franklin Yang (September 2021 – August 2022)
- Sofi Sinozich (September 2020 – August 2021)

Undergraduate Capstone Projects

Faculty Mentor

- Pragma Kumar, Orla Collins, Giulia Hoorens van Heyningen (2026) *Assessing the Viability of Weighting Methods for Hybrid Sampling Design*
- Danny Pham, Tariq Witherspoon, Andrew Shin (2026) *Machine Learning Model Selection with Complex Sample Survey Data*

Service

- Faculty Coordinator, Joint Program in Survey Methodology Short Course Program
- Co-Chair, Social Data Science Program Committee
- Co-Chair, Department of Psychology Lecturer Search Committee
- Chair, Social Data Science Curriculum Committee
- Chair, Search Committee for Executive Director of the Center for Advances in Data and Measurement
- Member/Chair, JPSM/MPSM Masters Admissions Committee
- Member, Department of Economics Lecturer Search Committee
- Member, Department of Criminology and Criminal Justice Lecturer Search Committee
- Member, College of Information Studies Lecturer Search Committee
- Member, Social Data Science Advisor Search Committee
- Member, Social Data Science Steering Committee
- Member, JPSM/MPSM Teaching Assignment Committee

Skills

Statistical Software

Highly skilled in R, SQL and Python, including various packages such as Rcpp, shiny, caret, tidyverse, tidymodels, parallel, snow, and more (in R); pandas, scikit-learn, nltk, tensorflow, keras, and more (in Python); Tableau

Statistics

Highly skilled in a variety of methods, including but not limited to hypothesis testing, regression, mixed models, clustering analysis, social network analysis, social network models, network sampling methods, Monte Carlo simulation, Bayesian models, machine learning, natural language processing, web scraping, neural networks, parallel computing, and more.

Other Software

Skilled or proficient in the use of many other programs, including, but not limited to, Microsoft Word, Excel and Powerpoint; LaTeX.